

Onion



Potato

Carefully remove a fresh carrot plant from the soil and observe it. Look at the part we usually consume as 'carrot vegetable'. It is not a unripe fruit, but the tap root of the carrot plant. We can see that the tap root of the carrot is swollen. In the case of the carrot plant, the tap root has a different characteristics than the usual plants. Normally, each plant organ originally evolves to meet certain needs of the plant. For example, roots evolve primarily to anchor the plant and also to absorb water and mineral nutrients from the soil.

Leaves are adapted to optimize photosynthesis. Stems evolve to reach out to sunlight and also to conduct water from roots to leaves. However in certain plant species, specific parts have evolved further in unusual and surprising ways to meet certain other specific needs, In some plants, root, stem, and leaves change their shape and structure to perform special functions like storage of food, mechanical support, protection and other vital functions. This is known as modification.

What appear as the 'leaf' of a cacti are actually their stem and what appear as 'spine' on them are actually leaf. Its leaves are modified into spines, an adaptation to reduce transpiration. Photosynthesis is performed by the stem part of the plant. In this section let us study about the modification of root, stem and leaves.

5.4.1 Modification of Root

a. Roots for storage

Look at radish, turnip, beet root, and carrot. They all grow under the soil. As soon as you pluck it from the ground if you wash them gently, you will notice small roots dangling from their surface. All these vegetables are in fact roots of the plant. Instead of thin slender roots, they have become a place to store the food produced by them. Hence, they are thick and swollen. One can notice that the tap root of radish is in the shape of spindle, swollen in the middle and tapering at both ends. Such type of modified roots are called spindle shaped root.



Radish

At times, like in the case of turnip and beet root, the tap root can acquire a shape of top, that is spherical at the base and tapering shortly towards the apex. They are called as top shaped root.



Beet Root

In case of **carrot,** the shape is conical, broad at the apex and tapering gradually towards the base and such modified roots are called conical shaped root.



Carrot

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